Every formula has a formation sequence

The Open Logic Project contributors

openlogic:first-order-logic/syntax-and-semantics:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument:

Editorial note. The label prop:formed names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Induction on formulas. Same unit, the principle of induction on formulas. Closing line of the proof (source line 1265): by the principle of induction on formulas, every formula has a formation sequence.

Proposition 0.1. *Every formula A in* $\mathrm{Frm}(\mathcal{L})$ *has a formation sequence.*

Proof. Suppose A is atomic. Then the sequence $\langle A \rangle$ is a formation sequence for A. Now suppose that B and C have formation sequences $\langle B_0, \ldots, B_n \rangle$ and $\langle C_0, \ldots, C_m \rangle$ respectively.

1. If $A \equiv \neg B$, then $\langle B_0, \ldots, B_n, \neg B_n \rangle$ is a formation sequence for A.
2. If $A \equiv (B \wedge C)$, then $\langle B_0, \ldots, B_n, C_0, \ldots, C_m, (B_n \wedge C_m) \rangle$ is a formation sequence for A.
3. If $A \equiv (B \vee C)$, then $\langle B_0, \ldots, B_n, C_0, \ldots, C_m, (B_n \vee C_m) \rangle$ is a formation sequence for A.
4. If $A \equiv (B \to C)$, then $\langle B_0, \ldots, B_n, C_0, \ldots, C_m, (B_n \to C_m) \rangle$ is a formation sequence for A.
5. If $A \equiv (B \leftrightarrow C)$, then $\langle B_0, \ldots, B_n, C_0, \ldots, C_m, (B_n \leftrightarrow C_m) \rangle$ is a formation sequence for A.
6. If $A \equiv \forall x\, B$, then $\langle B_0, \ldots, B_n, \forall x\, B_n \rangle$ is a formation sequence for A.
7. If $A \equiv \exists x\, B$, then $\langle B_0, \ldots, B_n, \exists x\, B_n \rangle$ is a formation sequence for A.

By the principle of induction on formulas, every formula has a formation sequence. □